FELIX NATANAEL BUTARBUTAR

Software Engineer • AI Engineer

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SUMMARY

Software Engineer and Information Technology graduate with hands-on experience developing enterprise systems and full-stack applications, combined with a strong specialization and passion for Artificial Intelligence. Currently engineering enterprise CRM solutions using Microsoft Dynamics 365 and MSSQL at PT Bintang Aplikasi Perkasa. Proven track record delivering production web architectures and AI pipelines across computer vision, NLP, and LLM evaluation for enterprise clients, bridging robust software engineering with applied AI innovation.

EXPERIENCE

PT Bintang Aplikasi Perkasa — **August 2026 – Present**
Software Engineer — South Jakarta, DKI Jakarta

- Develop and customize enterprise CRM solutions using **Microsoft Dynamics 365** and **.NET (C#)**, implementing custom plugins and backend services to streamline business workflows.
- Design, optimize, and maintain relational database schemas, complex queries, and stored procedures using **Microsoft SQL Server (MSSQL)** for robust data transactions.
- Build automated workflows and AI pipelines using **n8n**, integrating **Whisper** for audio transcription and **LLMs** for automated interaction summarization and CRM ticket processing.

INaAI — **February 2026 – August 2026**
AI Engineer Intern — Central Jakarta, DKI Jakarta

- Prepared computer vision training datasets using CVAT and performed homography calibration across 58 CCTV cameras for a retail shopper analytics system (PT Mayora Indah Tbk), mapping pixel coordinates to store floor plan coordinates.
- Conducted adversarial testing, prompt injection testing, and hallucination evaluation for an AI-powered health triage chatbot (BPJS Ketenagakerjaan), including LLM benchmarking and structured test case design.
- Performed multi-domain data sourcing and exploratory data analysis for a cocoa commodity price prediction model, consolidating variables across market pricing, climate anomalies, geopolitical risk, and agricultural production.

PT Aoki Mori Indonesia — **April 2025 – December 2025**
Full Stack Developer — Siborongborong, North Sumatra

- Engineered a full-stack academic management platform for LPK SO Mori Centre using **Go (Gin), GORM, JWT**, and **Next.js 16 / TypeScript**, covering 30+ RESTful API endpoints for class enrollment, assignment submission, and auto-graded online examinations.
- Developed a company profile website integrated with an online student registration system and multilingual support (**Japanese, English, Indonesian**).
- Containerized application services using **Docker Compose** and delivered responsive UI aligned with institutional branding.

EDUCATION

Binus University — **September 2026 – September 2028 (Expected)**
Bachelor of Computer Science — Jakarta, Indonesia

Del Institute of Technology — **September 2023 – August 2026**
Associate Degree in Information Technology • GPA: 3.50/4.00 — Toba, North Sumatra

PROJECTS

RuangRPP – AI-Assisted Lesson Plan & Curriculum Platform
Full-Stack Developer — August 2026

- Architected and developed a full-stack platform using **Next.js (React 19, TypeScript)** and **FastAPI (Python)** with **Docker Compose** and **PostgreSQL 17** for curriculum knowledge management.
- Built a secure file ingestion pipeline for PDF and DOCX documents with anti-malware safeguards (anti-zip bomb, sub-process bounds), extracting structured tables and text across 200+ pages.
- Implemented **Argon2id** session authentication, RBAC, request throttling, and automated rule-based curriculum validation (Phases A–C) for citation-backed **RAG / LLM** lesson planning.

Technologies: Next.js (React 19), FastAPI, PostgreSQL 17, Docker Compose, Python, TypeScript, SQLAlchemy, Pytest

GlitchRun – Real-Time Multiplayer Crash Game Platform
Software Engineer May 2026

- Architected a high-concurrency real-time game engine in Go using goroutines, channels, and a synchronized WebSocket Hub, broadcasting sub-100ms game-loop ticks to multiple concurrent clients.
- Implemented a Provably Fair cryptographic verification system using HMAC-SHA256 and SHA-256 pre-commitments, enabling transparent and mathematically verifiable random outcome generation.
- Eliminated race conditions and double-spend vulnerabilities by designing an atomic cashout mechanism with custom Redis Lua scripts for sub-millisecond bet validation and asynchronous PostgreSQL persistence.
- Containerized a 4-tier microservices architecture (Go, Next.js, Redis, PostgreSQL) using multi-stage Docker Compose for consistent local development and reproducible deployment.

Technologies: Go (Golang), Next.js 14, TypeScript, Redis, PostgreSQL, WebSockets, Docker, Zustand, TailwindCSS

Implementation of Whisper ASR and DeBERTa v3-Base for Evaluating Coherence of IELTS Speaking Answers
Machine Learning Engineer August 2025 – February 2026

- Developed a web-based automated IELTS Speaking evaluation system integrating **Whisper** for automatic speech recognition and **DeBERTa v3-Base** for evaluating semantic coherence between prompts and responses.
- Developed machine learning components using **Python, PyTorch, and Hugging Face Transformers**, including data collection and model integration with application frameworks.
- Built a **FastAPI** backend for model serving and integrated the inference pipeline with a **Next.js** frontend via REST API.

Technologies: Python, PyTorch, Hugging Face Transformers, Whisper, DeBERTa v3-Base, FastAPI, Next.js, REST API, Google Colab

Student Stress Level Prediction Using Ensemble 1D-CNN and XGBoost
Machine Learning Engineer August 2025 – December 2025

- Developed an ML classification system to predict student stress levels (**Distress, Eustress, No Stress**) from questionnaire data, applying feature engineering and **SMOTE** balancing.
- Designed and trained a **2-block 1D-CNN** for feature extraction and integrated learned embeddings with an optimized **XGBoost** classifier.
- Evaluated model performance with confusion matrices and deployed an interactive **Streamlit** application with CPU/GPU execution support.

Technologies: Python, TensorFlow, Keras, XGBoost, Pandas, NumPy, Streamlit, Flask

IT Del Student Information Management System
Full-Stack Developer January 2025 – July 2025

- Developed a web-based academic information management system with 24 functional features using **Laravel, Inertia.js, React, and PostgreSQL**, implementing role-based access control (RBAC) and secure authentication across 4 system roles.

Technologies: Laravel, Inertia.js, React, PostgreSQL

TECHNICAL SKILLS

Programming Languages:	Python, Golang, C# (.NET), JavaScript, PHP, Dart
AI / Machine Learning:	Machine Learning, Deep Learning, NLP, Automatic Speech Recognition (ASR), Computer Vision, LLM Evaluation, Whisper, DeBERTa, XGBoost
AI / ML Frameworks:	PyTorch, TensorFlow, Keras, Hugging Face Transformers
Web & Application:	FastAPI, Next.js, React.js, Laravel, Flutter, Streamlit
Databases:	Microsoft SQL Server (MSSQL), PostgreSQL, MySQL, Qdrant
Agentic AI & Coding Tools:	Agentic AI, Claude Code, OpenAI Codex, OpenCode
Tools & Technologies:	Microsoft Dynamics 365, n8n, CRM Systems, REST APIs, Model Deployment, CVAT, Google Colab, Firebase

ACHIEVEMENT

Finalist, DEL AI Hackathon (Nasional) 2026
Institut Teknologi Del

3rd Place, Coding Day Competition 2025
Faculty of Vocational Studies, IT Del

TRAINING & COURSES

AWS Machine Learning Specialty – Hands On! 2025
Udemy